

HUSSEIN ELSAYED

Unity Developer | XR / VR | Multiplayer | EdTech

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Open to remote, hybrid & on-site roles internationally · Available to relocate

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Portfolio: <https://husseinelayed.github.io/>

PROFESSIONAL SUMMARY

Unity Developer with professional experience building **multiplayer games, VR/XR applications, AR experiences, robotics simulations, educational technology platforms, interactive training systems, and mobile applications**. Strong experience in **C#, Unity, multiplayer and client-server architecture, VR/XR, AR, UI/UX, backend integration, REST APIs, visual programming, custom editors, Python code generation, AI/computer vision, and third-party SDK integration**. Experienced in developing complete interactive systems from **software architecture and UI/UX design to gameplay, networking, authentication, platform services, and performance optimization**. Educational technology experience includes developing robotics programming and simulation systems and teaching programming through **Scratch, Blockly, Python, JavaScript, React, Flutter, C#, Unity, and robotics**.

PROFESSIONAL EXPERIENCE

Kid Koder Academy

Instructor — Programming, Game Development & Robotics

Aug 2026 – Present

- Teach programming **from scratch**, starting with computational thinking and beginner-friendly visual programming.
- Teach **Scratch and Blockly**, including variables, conditions, loops, events, operators, functions, and programming logic.
- Teach students how **Scratch-style block editors work**, including concepts behind visual programming and custom block-based programming environments.
- Guide students through building games from scratch, including **Pong, multiplayer Pong, Super Mario-style games, and car/racing games**.
- Use game development to teach programming, problem-solving, algorithms, and logical thinking.
- Progress students from visual programming into text-based programming, including **Python and JavaScript**.
- Introduce modern development technologies through practical projects, including **React and Flutter**.
- Teach **C# and Unity** for students progressing toward professional game development.
- Teach robotics and programming concepts through hands-on projects and interactive simulations.
- Adapt lessons for students with different programming experience levels.

Freelance / Independent Unity Developer - [EgyGames](#)

2026 – Present

Sigato V2 — Multiplayer 2D Board Game

- Continued development of **Sigato V2**, expanding the original multiplayer game developed at DGSs.
- Designed and implemented the game's **UI/UX independently using Figma**, translating designs into Unity UI.
- Designed and implemented the project's software architecture using **Extenject / Dependency Injection**.
- Extended multiplayer architecture with **client-server synchronization, online communication, lobby, and matchmaking systems**.
- Integrated **Photon PUN 2** for multiplayer networking.
- Integrated **Unity Gaming Services (UGS), Firebase, and Google Sign-In**.

- Integrated **Google Play Games and Game Center**.
- Implemented **Analytics, In-App Purchases, Leaderboards, Achievements, Advertising, and localization**.
- Implemented **English, Arabic, and French localization**.
- Developed **daily rewards, challenges, shop, rewards, themes, audio, settings, and reusable popup systems**.
- Used **UniRx** and **LitMotion** for reactive systems and responsive UI interactions.
- Optimized the application for **mobile and cross-platform deployment**.

EKSON Technology / AlMotahida — [SciVR](#)

Unity Developer — VR / XR / Educational Applications

Sep 2024 – Jul 2026

- Developed **VR/XR, educational, medical, scientific, and interactive Unity applications**.
- Developed multiple VR projects using **VRIF (VR Interaction Framework)**.
- Built and optimized VR applications for **Meta Quest and Pico** standalone headsets.
- Developed and deployed **WebGL** interactive experiences for browser-based platforms.
- Implemented immersive VR interactions, object manipulation, UI, navigation, animation, audio, and interactive workflows.
- Developed VR virtual tours including **Pizza Tower and Acropolis** experiences.
- Developed medical and dental VR applications including **Dentist, Dentist Hygiene, Skin Care**, and related training experiences.
- Contributed to **5+ Mathematics, 5+ Biology, and 5+ CTE** educational projects.
- Optimized applications for standalone VR hardware and WebGL deployment.

DGSs

Game Developer — Sigato V1

Jul 2024 – Aug 2024

- Programmed and launched **Sigato**, a multiplayer 2D board game deployed directly to the Google Play Store.
- Implemented **real-time client-server synchronization**.
- Built **lobby and matchmaking pipelines**.
- Implemented online communication states and multiplayer gameplay synchronization.
- Established the initial gameplay, UI, and networking architecture that became the foundation for **Sigato V2**.

Thaka (Thaka international) / BE-STEAM (prev name) — [01Skills](#)

Unity Developer / Robotics & EdTech Developer

2023 – 2024

- Contributed to [01Skills](#), an educational technology platform serving thousands of students/users through interactive coding, robotics simulation, learning, practice, and competition.
- Developed and enhanced robotics simulation supporting **LEGO EV3 and SPIKE-style programming**.
- Extended block-based programming with **loops, conditions, logic, comparisons, arithmetic, variables, robot movement, sensors, and control structures**.
- Implemented support for **complex nested mathematical and logical expressions**.
- Developed comprehensive **EV3 and SPIKE block-programming systems**.
- Built a custom programming editor supporting **Block-to-Python and Code-to-Blocks** workflows.
- Implemented project loading and saving and a portable **Python code-generation workflow**.
- Connected programming blocks with simulated robot behavior, robotics mission logic, and challenges.
- Worked on the **Panama educational robotics mission**, including robot simulation, block programming, mission objectives, and execution logic.

PSTC — Saudi Arabia

Game Developer

Oct 2021 – Apr 2022

- Engineered a **multiplayer gamified mobile application** integrated into the InsuranceMarket platform.
- Implemented real-time multiplayer functionality using the **AppWarp multiplayer networking plugin**.

- Built multiplayer rooms, real-time communication, and synchronized game states.
- Built secure **login/sign-up authentication, user profiles, guest sessions, and real-time live chat rooms.**
- Integrated **Unity Ads and Start.io** SDK networks.
- Performed **texture compression, script refactoring, and code profiling** to improve application size and runtime performance.
- Optimized assets, scripts, rendering, and runtime performance for mobile devices.

FREELANCE PROJECTS

VR Clean Desk Policy

Interactive VR Training Simulation

- Developed a Unity-based VR workplace training simulation.
- Implemented **VR interactions, object manipulation, task-based workflows, and spatial audio/narration.**
- Developed the experience for standalone VR platforms.

GetMoney

Economic Simulation Game — Unity / PHP / MySQL

- Developed gameplay systems using **Unity, PHP, and MySQL.**
- Built custom backend functionality through **REST APIs.**
- Connected Unity gameplay systems with server-side data.
- Implemented interactive economic and simulation mechanics.

SELECTED PROJECTS

AR Card Gameplay System

Augmented Reality / Unity

- Developed an AR tracking system mapping **physical cards and anchors to virtual interfaces and gameplay elements.**
- Implemented image/marker tracking to recognize physical cards and connect them with corresponding digital interactions.
- Created an interactive bridge between **physical objects and virtual Unity gameplay/UI systems.**

AI & Computer Vision

- Developed Unity-based AI and computer vision experiences using **Unity Sentis, Unity ML-Agents, AI Behavior Trees, and object detection/recognition.**

Cinema Cars Race

- Developed a Unity racing experience involving gameplay systems, vehicles, UI, and interactive environments.

Unity Matching Game

- Developed a matching/puzzle game with gameplay, UI, progression, rewards, and interactive systems.

TECHNICAL SKILLS

Programming Languages

C#, Python, Java, C++, JavaScript, PHP, HTML5

Unity & Game Development

Unity, Unity 6, 2D, 3D, Gameplay Programming, UI/UX, Physics, Animation, Simulation, Custom Editors

Multiplayer & Networking

Client-Server Architecture, Multiplayer, Photon PUN 2, AppWarp, Mirror, WebSockets, Lobby Systems, Matchmaking, Real-Time Synchronization

XR / VR / AR

VR, XR, AR, VRIF, OpenXR, XR Interaction Toolkit, Meta Quest, Pico, Vuforia, VR Training, Virtual Tours

Backend & APIs

PHP, REST APIs, GraphQL, Firebase, MySQL, MongoDB, MongoDB Realm

Architecture & Software Engineering

OOP, SOLID, Design Patterns, Dependency Injection, Extenject, Event-Driven Architecture, Service Locator, Modular Systems, Data Structures & Algorithms, Performance Optimization

UI/UX & Design

Figma, Unity UI, UI/UX Design, Prototyping, Responsive UI

Visual Programming & Robotics

Scratch, Blockly, Block-Based Programming, LEGO EV3, LEGO SPIKE, Visual Programming, Python Code Generation, Block-to-Code, Code-to-Blocks, Compiler Pipelines, Custom Programming Editors

AI & Computer Vision

Unity Sentis, Unity ML-Agents, AI Behavior Trees, Computer Vision, Object Detection

Platform Services & SDKs

Firebase Authentication, Google Sign-In, Analytics, In-App Purchases, Unity Ads, Start.io, Leaderboards, Achievements, Google Play Games, Game Center

Other Technologies

React, Flutter

Platforms

Android, iOS, PC, WebGL, Mobile, Meta Quest, Pico

Tools

Git, GitHub, Jira, Visual Studio, Android Studio, Blender, Photoshop, Figma

LANGUAGES

Arabic: Native

English: Professional Working Proficiency